

Reinforcement Learning-Based Control for the ROV

Introduction

In addition to the imitation learning approach, a Reinforcement Learning (RL) framework was incorporated to explore autonomous navigation capabilities for the Remotely Operated Vehicle (ROV). Reinforcement Learning enables an agent to learn optimal control policies by interacting with the environment and maximizing a reward signal. Unlike supervised learning methods such as behavior cloning, RL allows the system to improve its behavior through experience and feedback.

The RL controller operates on the onboard computer and communicates with the ESP32 microcontroller, which manages the low-level hardware interface including sonar sensors and thruster control.

Reinforcement Learning Concept

Reinforcement Learning is a machine learning paradigm in which an **agent interacts with an environment by taking actions and receiving feedback in the form of rewards**. The objective of the agent is to learn a policy that maximizes the cumulative reward over time.

The RL process is defined by the following components:

- **State (Observation)**
The state represents the current perception of the environment. In the ROV system, the state consists of the sonar distance measurements:
 - Front distance
 - Left distance
 - Right distance
- These measurements provide information about nearby obstacles and free space.
- **Action**
The action defines the control command generated by the RL policy. In the ROV system, the action consists of two continuous control variables:
 - Forward velocity
 - Steering command
- **Reward**
The reward evaluates the quality of the action taken by the agent. The reward function encourages safe navigation and forward movement while discouraging collisions or unsafe behavior.

- **Policy**
The policy is a neural network model that maps sensor observations to control actions.

Reinforcement Learning Algorithm

The RL controller used in this system is based on the **Proximal Policy Optimization (PPO)** algorithm. PPO is a policy-gradient reinforcement learning method designed for stable and efficient training in continuous control environments.

PPO is widely used in robotics because it provides:

- Stable policy updates
- Efficient training with neural networks
- Good performance in continuous control tasks
- Robustness against large policy updates

The PPO algorithm trains a neural network policy that outputs control commands directly from sensor observations.

Pretrained Reinforcement Learning Model

Instead of training an RL model entirely from scratch, a pretrained reinforcement learning model was obtained from the **Hugging Face Model Hub**. The selected model is:

ARG-NCTU/hrl-ppo-usv

This model was originally developed for autonomous control of an **Unmanned Surface Vehicle (USV)** using hierarchical reinforcement learning and the PPO algorithm.

Although the original model was designed for surface vehicle navigation, it provides a useful baseline architecture for marine robotics applications involving continuous control and obstacle avoidance.

The pretrained model was implemented using the **Stable-Baselines3** reinforcement learning framework.

Model Architecture

The pretrained policy consists of a neural network that receives the environment state as input and outputs continuous control actions.

The input to the neural network includes:

- Distance to the obstacle in front of the vehicle
- Distance to the obstacle on the left side

- Distance to the obstacle on the right side

The output of the policy network includes:

- Forward motion command
- Steering command

The neural network learns a nonlinear mapping between sensor measurements and control actions that allow the vehicle to navigate safely.

Integration with the ROV System

The reinforcement learning controller runs on the onboard computer and communicates with the ESP32 microcontroller through a serial communication interface.

The interaction process follows the steps below:

1. The ESP32 reads distance measurements from the sonar sensors.
2. The sonar measurements are transmitted to the PC through the serial interface.
3. The RL model processes the sensor data and computes a control action.
4. The PC sends the computed velocity and steering commands back to the ESP32.
5. The ESP32 converts the commands into PWM signals for the thrusters.

This process forms a continuous perception–decision–action loop.

Safety Layer

To ensure safe operation during experimentation, a safety mechanism was implemented between the RL controller and the thrusters.

The safety layer monitors the sonar measurements and overrides the RL action if an obstacle is detected within a critical distance threshold.

If the front sonar detects an obstacle that is too close, the safety system automatically stops forward motion and commands a turning maneuver to avoid collision.

This safety mechanism ensures that the ROV does not rely solely on the learned policy for critical decisions.

System Architecture

The complete control architecture consists of two main layers:

Low-Level Control Layer (ESP32)

- Sonar sensor acquisition
- Thruster control through PWM signals
- Communication with the PC
- Fail-safe protection

High-Level Control Layer (PC)

- Reinforcement learning policy execution
- Sensor data processing
- Navigation decision making

This separation of responsibilities allows computationally intensive algorithms to run on the PC while maintaining reliable real-time hardware control on the ESP32.

Advantages of the RL Approach

The reinforcement learning controller offers several advantages for autonomous navigation:

- Ability to learn complex control strategies
- Adaptation to dynamic environments
- Continuous control of speed and steering
- Potential for performance improvement through additional training

Compared to rule-based control methods, RL can discover more efficient navigation strategies that may not be explicitly programmed.

Limitations

Although reinforcement learning provides powerful control capabilities, several challenges remain:

- RL models require extensive training data.
- Policies trained in simulation may not transfer perfectly to real environments.
- Safety mechanisms are necessary during real-world deployment.

For this reason, the RL system in this project is combined with safety overrides and tested gradually.

Conclusion

The reinforcement learning module represents the high-level intelligence of the ROV control system. By integrating a pretrained PPO policy from the Hugging Face Model Hub with the ESP32-based hardware control layer, the system demonstrates the feasibility of applying modern reinforcement learning techniques to underwater robotic navigation.

The architecture allows further improvements through additional training, policy fine-tuning, and integration with more advanced perception systems.